



ANTIOXIDANT EFFECTS OF AQUEOUS GARLIC EXTRACT ON THE QUALITY AND FERTILIZING POTENTIAL OF LIQUID-PRESERVED BOAR SEMEN

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(AGE),
Sperm fertilizing efficiency,
sperm quality parameters,
liquid storage,
Lipid Peroxidation.

ABSTRACT

Efficient utilization of artificial insemination improves livestock genetic profile in the shortest possible time. For this technique to succeed, semen preservation without damaging spermatozoa's fertilizing ability is necessary. However, liquid storage of semen at 17 °C can reduce cellular integrity due to perceived oxidative activities through cellular metabolic process during storage. This study thus investigated the efficacy of Aqueous Garlic Extract (AGE) in maintaining boar sperm structural and functional integrity during semen extension and liquid storage. Five treatments (three replicates each) were tested: a control (Beltsville Thawing Solution (BTS) only + semen, 3:1 ratio) as treatment 1 and AGE-supplemented groups of 50, 75, 100, and 125 µL as treatment 2, 3, 4, and 5, respectively. Immediately, semen was mixed with the extender; quality assessments were carried out (0 h) and subsequently at 24, 48, and 72 hours of storage. Livability, pH, plasma membrane integrity (PMI), and lipid peroxidation (LP) were examined in a completely randomized design. Descriptive statistical methods were adopted to analyze the collected data, and ANOVA ($p < 0.05$). Result showed that Aqueous garlic extract supplementations on sperm did not negatively impact the entire semen variables assessed. Lipid Peroxidation effect on sperm varied significantly, however, semen supplemented with 100µL AGE inclusion at 48hour of liquid storage showed optimum protective effect against antioxidative stress. PH values hover within the ideal range and maintained higher liveability than the control up to 72 hours. Plasma Membrane Integrity decreased gradually over the storage period across the treatment groups. In conclusion, antioxidative properties were demonstrated by AGE, and it's capable of improving spermatozoa functional integrity under liquid storage.

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INTRODUCTION

Artificial insemination (AI) is among the viable reproductive biotechnologies adopted in modern swine production, serving as a catalyst for genetic enhancement, productivity gains, and herd health optimization (Waberski et al., 2018; Knox, 2016). Through the extensive use of genetically superior boars, through AI adoption, producers are encouraged to optimize genetic improvement, accelerate breeding programmes, mitigate the spread of reproductive diseases, thereby lowering the costs of maintenance of breeding males to sustain reproductive efficiency even under suboptimal environmental conditions.

Boar semen exhibits rare vulnerability for preservation. They are produced in large volumes and exhibit exceptional sensitivity to chilling injury, necessitating storage within a narrow thermal range of 15–18 °C (Henning et al., 2022). Prolonged storage inevitably results to reduction

of spermatozoa functional integrity and quality parameters due to successive oxygen radicals' generation from continuous sperm metabolism. Roca et al. (2006) established that semen successful preservation during storage depend largely on maintaining sperm motility, membrane integrity, and pH between 6.8 and 7.5; however, the polyunsaturated fatty acid-richness of boar sperm membranes renders them highly vulnerable to oxidative damage, thereby compromising fertilizing capacity (Aitken et al., 2016).

Conventional semen extenders provide a foundation for structural stability and pH buffering, yet they often fall short in combating oxidative stress during extended storage. The dependence on synthetic antioxidants poses additional concerns, as some were linked to toxicity and potential carcinogenic effects (Sanchez-Escalante et al., 2003). This fact intensified interest in safe, naturally derived

alternatives capable of providing efficacious antioxidative protection. Garlic (*Allium sativum*) is a strong natural antioxidant potential option. According to El-Saadony et al. (2024), garlic is rich in bioactive organosulfur compounds, particularly allicin, which exhibits potent antioxidant properties. Considering aqueous garlic extract (AGE) as a natural, affordable, and locally sourced additive may present an innovative method for semen preservation. Such innovation holds strong significance in resource-limited regions like Sub-Saharan Africa, where enhancing sperm quality and meeting the growing demand for animal protein require affordable and practical approaches (FAO, 2019).

MATERIALS AND METHODS

Experimental Location

The research was carried out at the University of Ibadan Teaching and Research Farm, Piggery Unit, Ibadan, Nigeria. While Semen sample collection was done in the farm, laboratory assessments were carried out in the Animal Physiology and Bioclimatology Laboratory, Department of Animal Science, Faculty of Agriculture, University of Ibadan. The study area lies within the tropical rainforest ecosystem at approximately 7°20'N latitude and 3°50'E longitude, at an elevation of approximately 277 m above sea level.

Experimental Animals

Healthy boars were fed with diets that met the animal nutrient requirements in quality and quantity. The animal was served with clean water throughout the course of the experiment and the pen sanitation was regularly done. The boars

were individually housed in different pens under the same environmental conditions.

Aqueous Extraction of Garlic (*Allium sativum* L.) procedure

Garlic bulbs used in this study were procured at a commercial market place in proximity. Having properly peeled, rinsed with distilled water and weighed. One hundred grams of peeled garlic cloves were homogenized in 200 mL of distilled water using an electric powered blender until uniform slurry was achieved. At first, the homogenate was filtered to remove large particles with the aid of clean muslin cloth and thereafter passed through Whatman filter paper to achieve a clear aqueous extract. The filtrate was then stored at approximately -1°C until usage.

Experimental Design and Treatments layout

Completely randomized design was employed in which semen samples were randomly allocated to five treatment groups with three replicates. Fresh semen collected was extended with Beltsville Thawing Solution (BTS) and supplemented with varying concentrations of aqueous garlic extract (AGE) as follows:

Treatment 1 (Control): 3 mL BTS with 1 mL of semen (without AGE)

Treatment 2: 3 mL BTS + 1 mL semen + 50 µL of AGE inclusions

Treatment 3: 3 mL BTS + 1 mL semen + 75 µL of AGE inclusions

Treatment 4: 3 mL BTS + 1 mL semen + 100 µL of AGE inclusions

Treatment 5: 3 mL BTS + 1 mL semen + 125 µL of AGE inclusions

Beltsville Thawing Solution was the semen extender used throughout the experiment.

Table 1: Composition of Beltsville Thawing Solution (BTS) Extender

Component	Quantity(g/L)
Glucose	37.00
Sodium citrate	6.00
Potassium chloride	0.75
EDTA	1.25
Sodium bicarbonate	1.25
Penicillin	1.10
Streptomycin	1.10

Source: Johnson (2000); EDTA: Ethylenediamine tetra acetate

Preparation of Boar for Semen Collection

The preputial area of each boar was carefully cleaned before semen collection to eliminate residues from urine, dirt, and other contaminated materials that could compromise semen quality. The long hair surrounding the preputial opening was meticulously trimmed to avoid contamination during collection.

Semen Collection, Dilution and Storage

The gloved-hand techniques semen collection method was adopted, and the sperm-rich only portion of the semen was retained. Immediately after collection, the semen was transferred into a pre-warmed insulated thermo flask that had been pre warmed to minimize thermal shock. The semen was subsequently diluted with Beltsville

Thawing Solution according to the designated treatment groups and stored at 17 °C in a thermoregulated refrigerator for evaluation over a 72-hour storage period. (Althouse et al., 1998). The required semen samples were removed from the regulated refrigerator and pre-warmed at 37 °C for 30 minutes before semen quality assessment at each evaluation period of 24, 48, and 72 hours.

Semen Evaluation

Semen quality assessments were carried out immediately after extension (0 h) and subsequently at 24, 48, and 72 hours of storage. Parameters evaluated included sperm livability, semen pH, plasma membrane integrity (PMI), and lipid peroxidation.

Sperm liveability

Assessment of spermatozoa liveability was done using the eosin–nigrosin staining technique. A single drop of semen was blended with eosin stain followed by nigrosin; a thin smear was prepared on a clean microscope slide after which it was left to dry naturally at room temperature. The prepared slides were subjected to light microscope examination at ×400 magnification. According to Crabo et al. (1972), live spermatozoa did not absorb the eosin stain whereas dead spermatozoa absorbed the stain. No fewer than 100 spermatozoa were assessed per slide, and live spermatozoa proportion was calculated and expressed in percentage.

Semen pH

The measure of acidity and alkalinity of a semen. The hydrogen ion concentration of each semen sample produced by spermatozoa metabolic activities during pre and post thaw motility was quantified using a calibrated digital pH meter (Mettler Toledo IP67®).

Determination of Lipid Peroxidation

The common method of determination of lipid peroxidation is by measuring malondyaldehyde (MDA) concentration using the thiobibaturic acid reactive substances (TBARS) assay, to know its extent of damage on cells. The major problem sperm quality suffers during storage is lipid peroxidation and due to boar sperm cell membrane high level of Poly Unsaturated Fatty Acid, they are vulnerable to Oxygen radicals, and in turn reduce spermatozoa viability and integrity. Following the classical protocol described by Buege and Aust (1978) with semen-specific adaptations (Suleiman et al., 1996) and aligned

with recent standardized procedures for TBARS/MDA measurement (Aguilar Diaz De Leon & Borges, 2020; Tsikas, 2017). A millilitre of seminal plasma was mixed with 2 mL of TBA-TCA reagent comprising 15% trichloroacetic acid, 0.375% thiobarbituric acid, and 0.25N hydrochloric acid. There was 30 minutes heating of the mixture in boiling water bath which was allowed to cool to room temperature before 10 minutes centrifugation at a speed of 3000 revolution per minute. Thereafter, the resultant supernatant was measured by spectrophotometer at 535 nm to estimate MDA levels.

Plasma Membrane Integrity

Sperm cell membrane integrity was examined using Hypo-osmotic swelling (HOS) test. The HOS solution consisted of 1.47 g sodium citrate and 2.70 g fructose dissolved in 200 ml of distilled water. An aliquot of 0.1 ml of semen was added with 1.0 ml of the HOS solution and incubated at 37 °C for 40 minutes, a drop of the mixture was therefore examined microscopically after incubation, and 100 spermatozoa were evaluated. Spermatozoa exhibiting tail swelling features were regarded as intact functional plasma membranes. (Muhammad et al., 2011).

Statistical Analysis

Descriptive statistical analysis was adopted for all experimental data followed by one-way analysis of variance (ANOVA) using SAS software (Version 9.2). Where significant treatment effects were detected ($P < 0.05$), treatment means with different superscript letters (a, b) were separated using Duncan's Multiple Range Test.

RESULTS

Characteristics of collected Fresh Boar Semen Prior to Extension

Table 2 explained the physical and microscopic characteristics of fresh boar semen collected prior extension. The average volume of the ejaculate was 242.80 mL, with a characteristic milky appearance with pH average of 7.24. Normal spermatozoa and Progressive sperm motility were both 95.00%, while mass activity was rated as +++. The semen temperature at 15 minutes after collection was measured to be 32.5 °C. All the value recorded was within the acceptable range for high quality boar semen and therefore considered suitable for extension and further usage.

Table 2: Semen characteristics of fresh boar ejaculate before extension

Parameters	Value
Volume (mL)	242.80
Colour	Milky

pH	7.24
Progressive motility (%)	95.00
Mass activity	++++
Normal Spermatozoa (%)	95.00
Temperature ¹ (°C)	32.5
1= Fifteen minutes after collection	

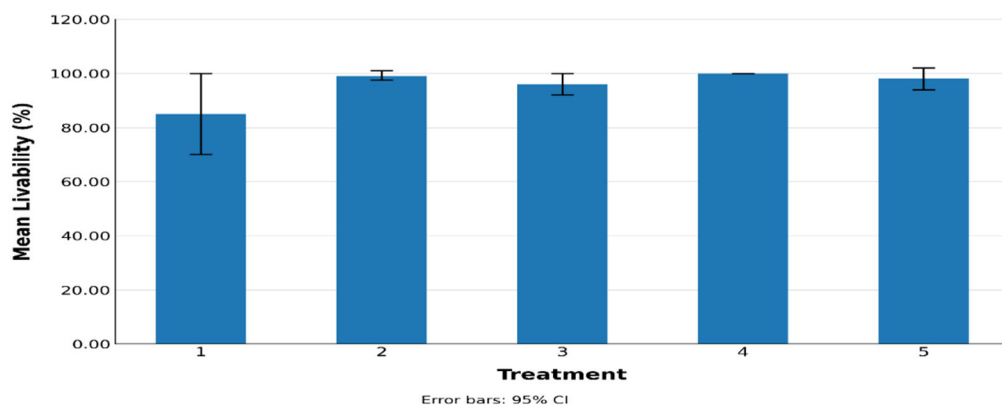


Figure 1: *Allium sativum* L aqueous extract effect on boar spermatozoa liveability (%) at initial (0 hour) of extension and storage

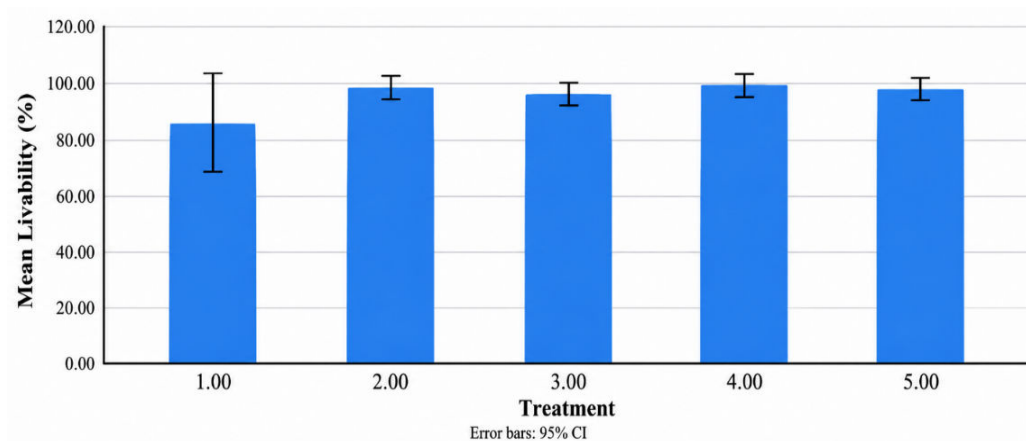


Figure 2: *Allium sativum* L aqueous extract effect on sperm liveability (%) in extended boar semen stored at 17°C semen at 24 hours of storage

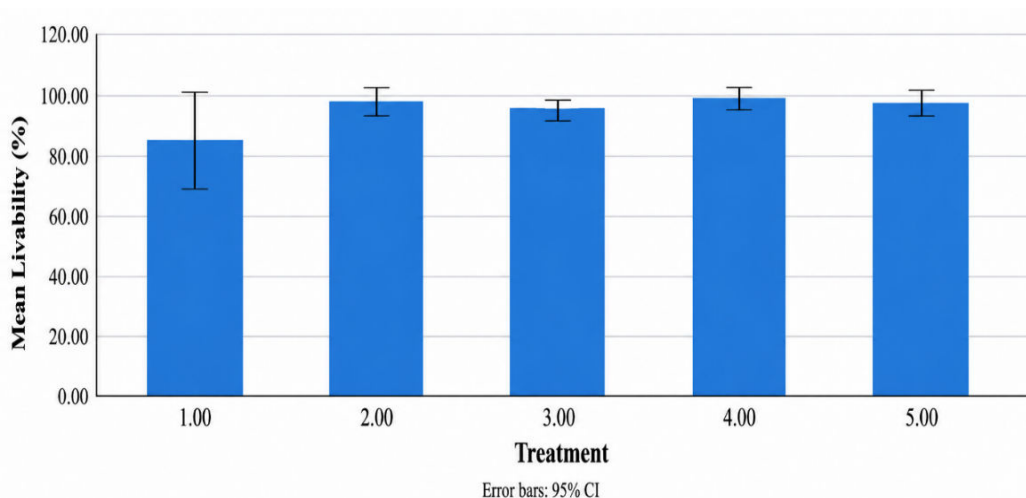


Figure 3: Effect of *Allium sativum* L aqueous extract on sperm liveability (%) in extended boar semen stored at 17°C at 48 hours of storage

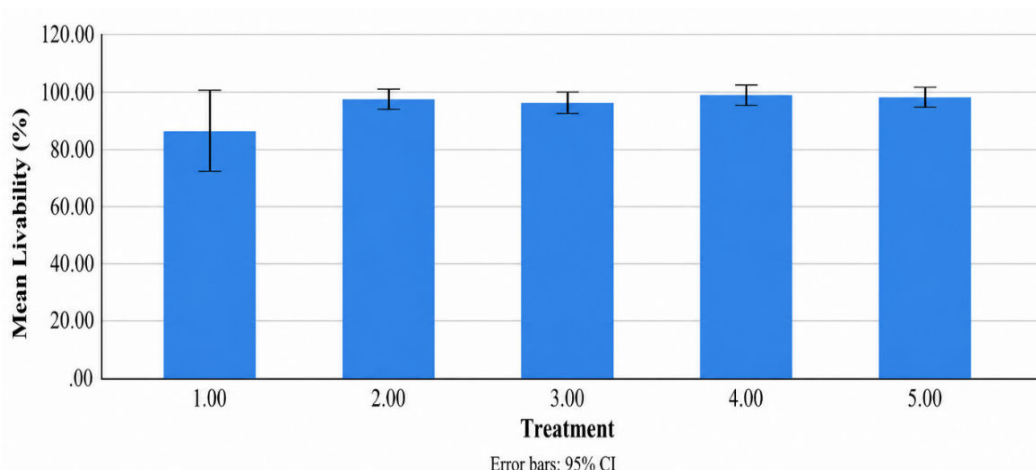


Figure 4: Liveability (%) of spermatozoa in extended boar semen stored at 17°C at 72 hours of storage under Aqueous extract *Allium sativum* L supplementation

Aqueous Garlic Extract inclusions effect on sperm liveability

Aqueous garlic extract (AGE) supplementations on sperm liveability during storage at 17°C is illustrated in Figures 1-4. Immediately after extension at (0 hr), considering sperm liveability, there was no observable difference among the treatment groups, although Treatment 3 (75 µL AGE inclusion) recorded the highest percentage of viable spermatozoa (99.33%), compared with 85.00% observed in the control group. At 24 hours of storage, (100 µL AGE supplementation)

maintained the highest percentage of viable sperm cells (96.67%) and differed significantly ($P < 0.05$) from the remaining treatments groups. At 48 hours, semen supplemented with 75 µL AGE (Treatment 3) exhibited the highest sperm liveability and remained significantly superior ($P < 0.05$) to the other treatment groups. Report in 72 hours of liquid storage refrigeration, sperm viability declined in all groups. However, semen treated with garlic extract maintained higher percentages of live spermatozoa against the control.

Table 3: Aqueous Garlic Extract supplementations effect on pH of Boar Semen

Storage (Hour)	Treatment					SEM
	1	2	3	4	5	
0	7.27	7.26	7.24	7.33	7.23	0.01
24	7.16	7.16	7.15	7.21	7.13	0.01
48	7.12	7.16	7.15	7.21	7.13	0.01
72	7.09	7.13	7.16	7.14	7.13	0.01

SEM: Standard Error of Mean

Aqueous Garlic Extract supplementations effect on Semen pH

Changes in semen pH throughout the preservation period are presented in Table 3. At the beginning of storage (0 hour), semen supplemented with 125 µL of AGE recorded the highest pH (7.33), whereas those that were treated with 100 µL of AGE produced the lowest (7.23). After 24 hours, the highest pH was observed in Treatment 4 (7.21), while Treatment 5 recorded the lowest value (7.13). A similar trend persisted at 48 hours,

with Treatment 4 maintaining the highest pH (7.21) and the control group exhibiting the lowest value (7.12). By 72 hours, the pH values ranged from 7.09 to 7.16. pH values for all treatments displayed a steady decline as storage progressed however, they maintained the normal physiological range considered suitable for boar semen preservation. Moreover, The numerical differences observed throughout the storage period were negligible.

Table 4: Effect of AGE on Lipid Peroxidation in Extended Boar Semen

Storage (Hour)	Treatment					SEM
	1	2	3	4	5	
0	0.65 ^a	1.78 ^c	0.82 ^a	1.38 ^{bc}	1.17 ^b	0.11
24	0.89 ^a	1.16 ^b	1.24 ^b	1.60 ^c	1.00 ^{ab}	0.70
48	0.73 ^b	1.31 ^c	0.62 ^b	0.43 ^a	0.59 ^{ab}	0.08
72	0.75 ^a	0.94 ^b	0.92 ^b	0.99 ^{bc}	0.80 ^a	0.44

a, b, c = Means within the same row with different superscripts differ significantly ($P < 0.05$); SEM: Standard Error of Mean

Effect of Aqueous Garlic Extract on Lipid Peroxidation

The effect of AGE supplementation on lipid peroxidation is summarized above. A significant treatment effect ($P < 0.05$) was observed during the storage period. At initial (0 hour), malondialdehyde (MDA) concentration in the control treatment group appeared low (0.65^a), followed closely by semen supplemented with 75 μ L of AGE (0.82), whereas Treatments 2 and 4

recorded comparatively a bit higher. After 24 hours, Treatments 5 (1.00^{ab}) exhibited the lowest lipid peroxidation values close to control T1 (0.89^a), while Treatment 4 showed the highest MDA concentration. At 48 hours, semen with 100 μ L AGE inclusion recorded the lowest lipid peroxidation (0.43), indicating reduced oxidative damage compared with the remaining treatment groups.

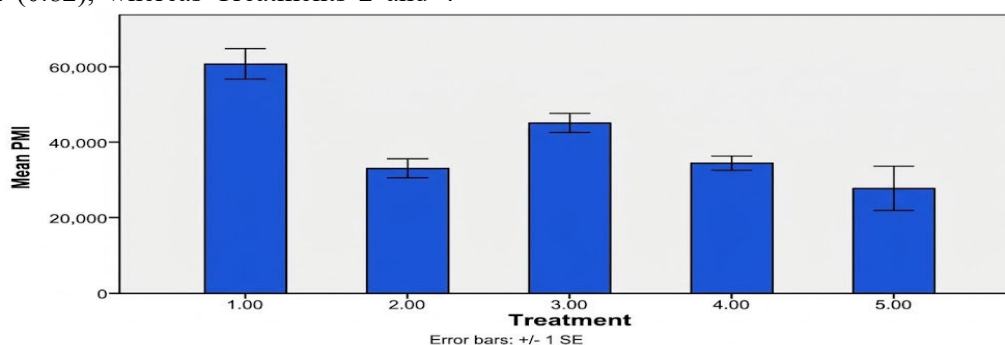


Figure 5: Effect of AGE on plasma membrane integrity (PMI) of sperm cell in extended porcine semen at 0 hour

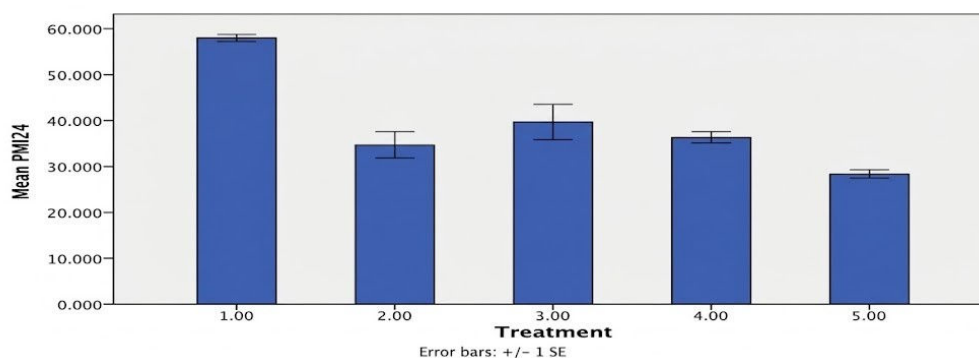


Figure 6: Effect of AGE on plasma membrane integrity (PMI) of spermatozoa in extended porcine semen at 24 hours of liquid storage

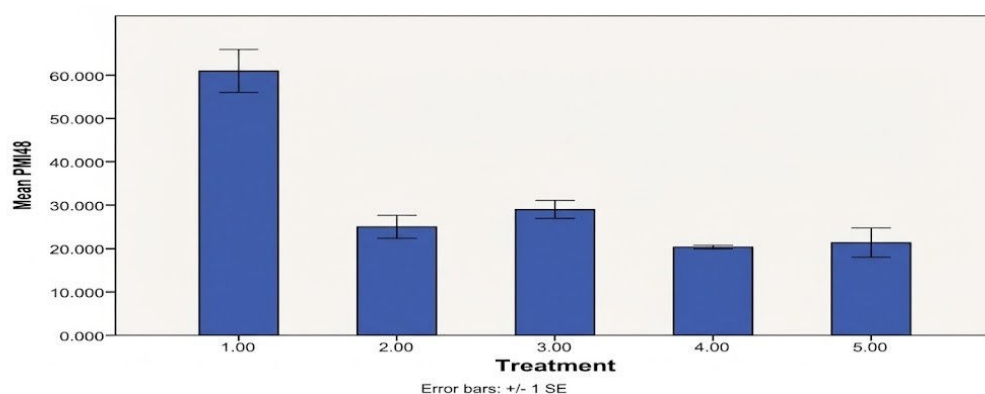


Figure 7: Aqueous Garlic Extract (AGE) effect on plasma membrane integrity (PMI) of sperm cell in extended porcine semen at after 48 hours liquid storage

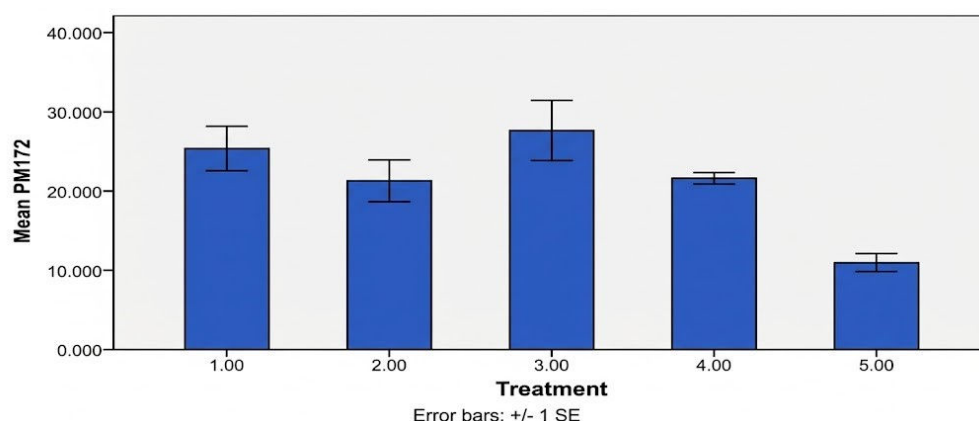


Figure 8: Aqueous Garlic Extract (AGE) effect on sperm cell plasma membrane integrity (PMI) in extended boar semen after 72 hours of storage

Aqueous Garlic Extract (AGE) supplementations effect on plasma membrane integrity (PMI) in extended boar semen

Plasma membrane integrity (PMI) after AGE supplementation during sperm liquid storage are presented in Figures 5-8. At 0 hours, significant differences ($P < 0.05$) were observed among all the treatments. The highest cell membrane integrity occurred in the control treatment group (60.67%), whereas 125 μg *Allium sativum* L. supplementation recorded the lowest value (27.70%). After 24 hours of storage, the control treatment spermatozoa membrane continued to maintain the best integrity, differing significantly ($P < 0.05$) from the AGE-treated groups. At 48 hours, plasma membrane integrity declined across all treatments however, the reductions were more pronounced in AGE-treated samples than in the control. After 72 hours of storage, there were considerable significant differences ($P < 0.05$) across the AGE supplemented treatment groups with 75 μg AGE inclusion treatment as the highest and 125 μg AGE supplemented group depicting the lowest cellular integrity.

DISCUSSION

Effect of Aqueous Garlic Extract on Sperm Livability

The present study demonstrated that supplementation of Beltsville Thawing Solution (BTS) with aqueous garlic extract (AGE) effectively maintained sperm viability during refrigerated storage. Although a gradual decline in livability was observed with increasing storage time, semen samples supplemented with AGE consistently exhibited higher percentages of viable spermatozoa than the untreated control, particularly at 24, 48, and 72 hours. These findings suggest that AGE contributed to preserving sperm cell survival throughout the storage period. The improved sperm livability observed in AGE-treated samples is most likely attributable to the antioxidant properties of garlic. El-Saadony et al. (2024) and Moretti et al. (2023) clarified that oxidative free radicals that occur in the process of sperm metabolism can be neutralized by component of bioactive organosulfur compounds which can be found in garlic. These compounds help preserve membrane functionality and mitigate premature sperm cell death during storage. The present findings are consistent with reports that

antioxidant supplementation enhances sperm survival during liquid preservation by protecting cellular structures against oxidative damage. Similar studies were conducted by Cerolini et al. (2000), Bucak et al. (2010), and Khalifa et al. (2021), who reported that natural antioxidants protect and improve sperm viability by reducing oxidative stress during semen storage. The sustained livability observed in the AGE-treated groups in this experiment therefore signified that aqueous garlic extract possesses sufficient antioxidant to serve as natural protective additive in boar semen extenders.

Effect of Aqueous Garlic Extract on Semen pH

The differences among treatments throughout the storage period from 0 to 72 hours at 17 °C storage temperature were recorded. The recorded pH value in all treatments from initial hour to 72 hours were close to the pH (7.24) of fresh semen collected, this propinquity could be as a result of pH of the semen and the extending medium being close (Rangsri et al., 2008). Although, Treatment 4 presented the highest value at 0 hour, however, it still falls within the normal pH range of viable semen (6.8 to 7.5). Rangsri et al., 2008 research output was in accordance with these findings stating that spermatozoa motility is often high in the pH range of 6.8 to 7.5. From 24 to 72 hours, there were differences in the pH mean value obtained in all the treatments however; all values hover within the acceptable range. This observation indicates that supplementation with aqueous garlic extract neither induced excessive acidification nor altered the buffering capacity of the extender. The ability of AGE-treated semen to maintain acceptable pH values suggests that garlic extract is compatible with Beltsville Thawing Solution and does not interfere with its buffering system. Serial reduction noticed in the pH value in all the treatments as storage progressed was expected, as is primarily associated with continuous sperm metabolism, during which glucose is converted into lactic acid and other metabolic by-products. Such gradual acidification has been widely reported during liquid semen preservation and reflects normal metabolic activity rather than deterioration of semen quality. This agrees with previous reports of Johnson et al., 2000 who stated that reduction in intracellular pH obviously slow down spermatozoa metabolism thereby increasing sperm cell survival during storage.

Effect of Aqueous Garlic Extract on Lipid Peroxidation of Extended Semen

Detection of malondialdehyde (MDA) provides an indication of cellular oxidation damage during semen storage. Lipid peroxidation is therefore known as one of the principal mechanisms responsible for deterioration of spermatozoa during storage. Cell inability to balance between reactive oxidative species (ROS) production and antioxidant material lead to cell lipid peroxidation and impair spermatozoa function (Shukla et al., 2012).

This occurs as a result of Boar sperm membranes that is rich in polyunsaturated fatty acids (PUFA) that are utmost vulnerable to oxidative invasion (Guthrie & Welch, 2007).

In this study, relatively low MDA concentrations were recorded throughout the storage period, with the lowest values observed in semen supplemented with 75 and 100 µL of aqueous garlic extract after 48 hours of storage. These findings indicated that Aqueous Garlic Extract inclusions actively reduced oxidation of sperm membranes, thereby limiting lipid peroxidation. Low values of LP recorded across the treatment period of 0 to 72 hours could first be attributed to the possible oxidative characteristics of spermatozoa and seminal plasma as backed by the findings of Zanella et al. (2016), who established that under normal physiological conditions, spermatozoa and seminal plasma are endowed with a multifaceted antioxidant defense system including enzymatic and non-enzymatic elements capable to protect the cell against ROS. The collection temperature and subsequent storage by refrigeration at 17 °C may also be a factor for the low value MDA achieved mostly across the treatments and this reflect Johnson et al., (2000), where he documented that two crucial elements for the storage of liquid boar semen are the storage medium's composition and the collection temperatures and storage.

According to Halliwell and Gutteridge (2015), antioxidants are compounds that prevent cells and biomolecules from oxidative damage by inhibiting or neutralizing the production of reactive oxygen species. The lowest values reported for malondialdehyde concentration in T3 and T4 at 48 hours and T5 at 72 hours of this study could be traced to specified concentration of Aqueous Garlic Extract (AGE) inclusion in the extender. Therefore, anti-oxidative expressions of garlic through its organosulfur constituents by scavenge of free radicals before they act on sperm cell membrane is noticeable.

Changes in Plasma Membrane Integrity of Boar Spermatozoa supplemented with AGE

This test is used to assess the semi-permeability of an intact cell membrane, spermatozoa swell when exposed to hypo osmotic medium, it is however an indication of functional membrane integrity (Zubair et al., 2015). Results obtained in this study demonstrated a significant decrease in the number of functional membranes integrity of boar spermatozoa preserved with AGE inclusion from 0 to 72 hours across the treatments. There was a general decline in plasma membrane integrity during the 72-hour storage period, meanwhile, some fluctuations were observed between individual storage intervals and treatments. The reduction in PMI on a general note at the extended storage period indicates progressive deterioration of spermatozoa membrane integrity during liquid preservation. Reductions in plasma membrane integrity during prolonged semen storage in this study is in agreement with the findings Karunakaran et al. (2017) and Zubair et al. (2015), who made a notable conclusion that membrane deterioration remains one of the principal limitations of liquid semen preservation despite antioxidant supplementation.

CONCLUSION

Significant antioxidative effects of aqueous garlic extract (AGE) were demonstrated during liquid storage of AGE supplement-BTS extended boar semen at 17 °C for as long as 72 hours. AGE preserved spermatozoa viability, pH stability, and minimal lipid peroxidation within the ideal physiological bounds. Due to its organosulfur bioactive chemical properties, AGE is ideal at reducing oxidative stress, as indicated by the reported decrease in malondialdehyde concentrations in AGE-treated samples. AGE's compatibility with common semen extenders are demonstrated by the fact that, despite a gradual loss in plasma membrane integrity, its inclusion had no negative consequences. These findings imply that AGE is a good natural substitute capable to replace artificial antioxidants for boar semen preservation.

RECOMMENDATION

To improve oxidative stability during storage, especially in production systems looking to substitute synthetic antioxidants, AGE could be added to boar semen extenders. Moreso, In vivo fertility outcomes, extended storage period beyond 72 hours, and proper inclusion rates should all be tested in future research.

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